

Project 4 Task 2

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Description:

My application takes a search string from the user and uses it to fetch and display food products and their nutritional values.

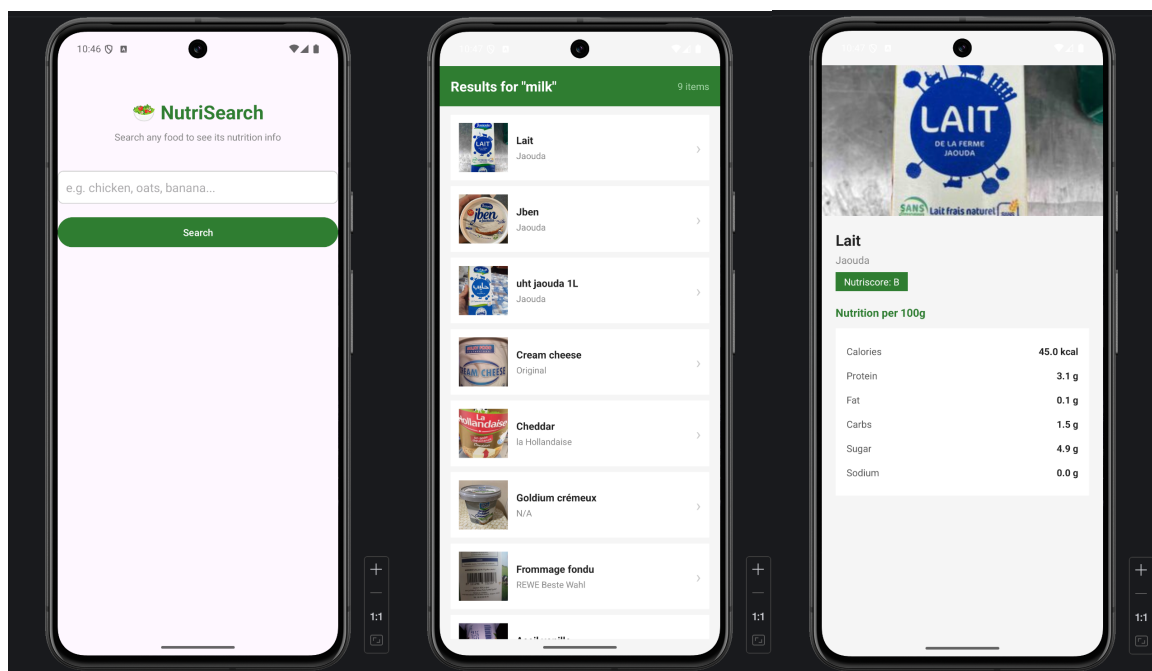
Here is how my application meets the task requirements

1. Implement a native Android application

The name of my native Android application project is NutriSearch.

a) Has at least three different kinds of views in your Layout

My application uses TextView, EditText, Button and ImageView



b) Require input from the user

Requires user to input a food item

c) Makes an HTTP request to your web service

My application makes HTTP request in ResultsActivity.java class to this URL:

scaling-waddle-q75vg74q9wwr3x4ww-8080.app.github.dev/api/seach?q=

d) Receives and parses XML or JSON formatter from the web service

Receives and parses JSON in ResultsActivity.java

e) Display new information

Included in about screenshot

f) Is repeatable

User can go back and search for a different food item without restarting the application

2. Implement a web application

The application is deployed on github codespaces

a) Using an HttpServlet to implement a simple API

Model: LogEntry

View: index.jsp

Controller: FoodSearchServlet

b) Receives an HTTP request from the native Android application

FoodSearchServlet receives an HTTP request with the parameter “q”

c) Executes business logic

Logs the request, calls the third party API, push the logs to database, returns the result

d) Replies to the Android application with a JSON formatter response

Yes it does, as mentioned earlier.

3. Handle error conditions

Handles missing parameter, product not found, retries 5 times for 503 error

4. Log useful information

Logs timestamp, query, client IP, response status, result count and duration

5. Store the log information in database

Connection string: "mongodb+srv://admin:changeme@ds-project-4.qbzgxy.mongodb.net/?appName=ds-project-4"

Shards:

ac-js7onao-shard-00-02.qbzgxy.mongodb.net:27017

ac-js7onao-shard-00-01.qbzgxy.mongodb.net:27017

ac-js7onao-shard-00-02.qbzgxy.mongodb.net:27017

6. Display operations analytics and full logs on a web-based dashboard:

08/04/2026, 23:06

NutriSearch Dashboard



NutriSearch Dashboard

Analytics

🔥 Top Search Query

coke

⚡ Avg Response Time (24h)

1596 ms

❌ Error Rate

50.0%

Top 5 Most Searched Queries

Query	Search Count
coke	12
test	5
banana	2
lasun	1
chicken	1

Recent Logs (last 50)

Timestamp	Query	Client IP	Status	Results	Duration
2026-04-09T01:51:17.690313462Z	banana	0:0:0:0:0:0:1	200	9	209 ms
2026-04-09T01:51:13.053971679Z	banaa	0:0:0:0:0:0:1	404	0	2925 ms
2026-04-09T01:51:00.771790045Z	lasun	0:0:0:0:0:0:1	404	0	1050 ms
2026-04-09T01:37:09.800618326Z	banana	0:0:0:0:0:0:1	200	9	1662 ms
2026-04-08T17:04:22.016068042Z	coke	0:0:0:0:0:0:1	200	10	7575 ms
2026-04-08T17:03:31.217497571Z	coke	0:0:0:0:0:0:1	200	10	2138 ms
2026-04-08T16:55:29.313978865Z	coke	0:0:0:0:0:0:1	500	0	5003 ms
2026-04-08T16:55:19.428362919Z	coke	0:0:0:0:0:0:1	500	0	5110 ms

https://scaling-waddle-q75vg74q9wwr3x4ww-8080.app.github.dev

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